

Homework 9

Limits at Infinity: Two Ways

Name: _____

Instructions

Use a **pen** and **show every step**. **Read through the readings slowly and ensure you understand everything**. Take your time and do not rush, it will only hurt you later. Take notes in your notebook of important information that you learn on here. This homework solves *one* limit two different ways — first by a quick intuition, then rigorously with the limit laws — so you can see that a good shortcut and careful algebra give the same answer. Read each section before its problems. You may use the internet + AI as a **teacher**, but do each problem yourself first and only use it once you fail.

The running example

Throughout this homework we will keep returning to one limit:

$$\lim_{x \rightarrow \infty} \frac{2x^2 + 3x - 1}{4x^2 - x + 5}.$$

Section 1 finds it in three lines by intuition. Section 2 finds it again, slowly and rigorously. Both should land on the same number.

Section 1 — Dropping the Lower-Order Terms

Reading — why the biggest term wins

When x is enormous, not all terms matter equally. Let's start with some intuition: suppose we have the expression $x + 1$. When x is a large number like 1,000 we have

$$x + 1 = 1000 + 1 = 1001.$$

Notice how the constant 1 contributed barely anything to our final value 1001. You can imagine as x gets really large like 1,000,000 the constant 1 does even less. So we can say mathematically, for large values of x ,

$$x + 1 \approx x.$$

Now let's compare x^2 and x at $x = 1,000,000$:

$$x^2 = 1,000,000 \times 1,000,000 = 1,000,000,000,000, \quad x = 1,000,000.$$

The x^2 term is a **million times larger** than x . The behavior you saw for the expression $x + 1$ also holds for variables. That is, for really large values of x ,

$$x^2 + x \approx x^2$$

Now looking at the expression $2x^2 + 3x - 1$, the $+3x$ and -1 terms are like fleas riding on an elephant — they are there, but they do not move the total in any noticeable way. So as $x \rightarrow \infty$,

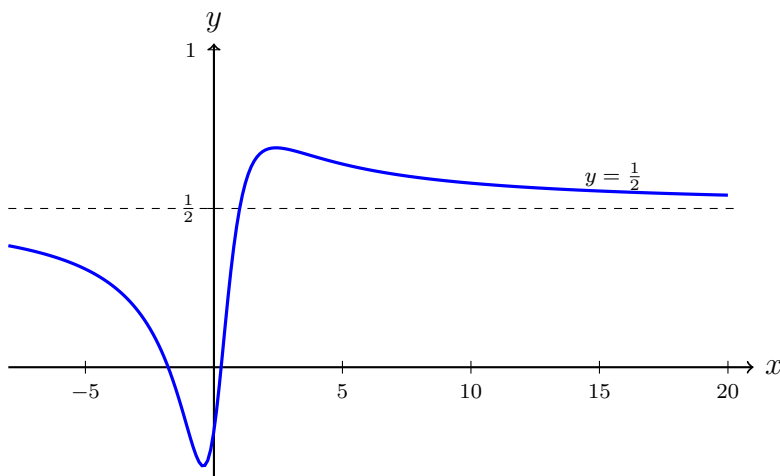
$$2x^2 + 3x - 1 \approx 2x^2, \quad 4x^2 - x + 5 \approx 4x^2.$$

The **highest-power term dominates**. Keeping only those dominant terms:

$$\lim_{x \rightarrow \infty} \frac{2x^2 + 3x - 1}{4x^2 - x + 5} \approx \lim_{x \rightarrow \infty} \frac{2x^2}{4x^2} = \lim_{x \rightarrow \infty} \frac{2}{4} = \frac{2}{4} = \frac{1}{2}.$$

The picture

Here is the graph of that function $f(x) = \frac{2x^2 + 3x - 1}{4x^2 - x + 5}$. Notice how, far out to the right, the curve flattens toward the dashed line $y = \frac{1}{2}$ — that horizontal line is exactly what “the limit at infinity is $\frac{1}{2}$ ” means.



Three possible outcomes

When dealing with a ratio of polynomials, comparing the dominant terms tells you which of three things will happen (ask yourself: do you know what a “*ratio of polynomials*” is? if not, search it up!):

1. **Same degree (numerator and denominator):** the limit is the ratio of the leading coefficients (our example: $\frac{2}{4} = \frac{1}{2}$).
2. **Degree of the denominator is greater:** the denominator grows faster, so the fraction is crushed to **0**.
3. **Degree of the numerator is greater:** the numerator grows faster, so the fraction blows up to $\pm\infty$ (no finite limit).

(ask yourself: do you know what a “*degree*” is? if not, you could not have understood the three bullet points above. search it up!)

Problems — use the dominant-term shortcut

For each, state which of the three cases it is (1, 2, or 3), then give the limit.

1. $\lim_{x \rightarrow \infty} \frac{5x^3 + 2x}{x^3 - 7}$

2. $\lim_{x \rightarrow \infty} \frac{3x + 4}{x^2 + 1}$

3. $\lim_{x \rightarrow \infty} \frac{x^2 - 6}{2x + 3}$

4. $\lim_{x \rightarrow \infty} \frac{7x^2 - x}{3x^2 + 2x - 8}$

Section 2 — Limit Properties and Algebra

Reading — the limit laws

The shortcut in Section 1 works because of a few precise rules. Suppose $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow \infty} g(x)$ both exist. Then:

Constant Law	$\lim c = c$
Sum Law	$\lim (f + g) = \lim f + \lim g$
Difference Law	$\lim (f - g) = \lim f - \lim g$
Constant Multiple Law	$\lim (c f) = c \lim f$
Product Law	$\lim (f g) = \lim f \cdot \lim g$
Quotient Law	$\lim \frac{f}{g} = \frac{\lim f}{\lim g} \quad (\text{provided } \lim g \neq 0)$

And the single fact that powers this whole topic — the one that makes “infinity” do useful work:

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \quad \text{for any } n > 0.$$

As x grows without bound, $\frac{1}{x}$, $\frac{1}{x^2}$, $\frac{1}{x^3}$, $\dots$ all shrink to 0. This is the tool that lets the algebra finish.

Worked example — the same limit, done rigorously

We evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 3x - 1}{4x^2 - x + 5}$ again. The trick: **divide every term, top and bottom, by the highest power present** — here x^2 .

$$\frac{2x^2 + 3x - 1}{4x^2 - x + 5} = \frac{\frac{2x^2}{x^2} + \frac{3x}{x^2} - \frac{1}{x^2}}{\frac{4x^2}{x^2} - \frac{x}{x^2} + \frac{5}{x^2}} = \frac{2 + \frac{3}{x} - \frac{1}{x^2}}{4 - \frac{1}{x} + \frac{5}{x^2}}.$$

Now apply the limit laws **one step at a time**, naming each one:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2 + \frac{3}{x} - \frac{1}{x^2}}{4 - \frac{1}{x} + \frac{5}{x^2}} &= \frac{\lim_{x \rightarrow \infty} \left(2 + \frac{3}{x} - \frac{1}{x^2}\right)}{\lim_{x \rightarrow \infty} \left(4 - \frac{1}{x} + \frac{5}{x^2}\right)} && \text{(Quotient Law)} \\ &= \frac{\lim_{x \rightarrow \infty} 2 + \lim_{x \rightarrow \infty} \frac{3}{x} - \lim_{x \rightarrow \infty} \frac{1}{x^2}}{\lim_{x \rightarrow \infty} 4 - \lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{5}{x^2}} && \text{(Sum \& Difference Laws)} \\ &= \frac{\lim_{x \rightarrow \infty} 2 + 3 \lim_{x \rightarrow \infty} \frac{1}{x} - \lim_{x \rightarrow \infty} \frac{1}{x^2}}{\lim_{x \rightarrow \infty} 4 - \lim_{x \rightarrow \infty} \frac{1}{x} + 5 \lim_{x \rightarrow \infty} \frac{1}{x^2}} && \text{(Constant Multiple Law)} \\ &= \frac{2 + 3(0) - 0}{4 - 0 + 5(0)} \\ &= \frac{2}{4} = \frac{1}{2}. \end{aligned}$$

Same answer as Section 1 — the shortcut and the rigorous method agree.

Problems — divide by the highest power, then use the laws

Show your work rigorously. That is, show the division and where the $\frac{1}{x^n}$ terms vanish. For each appropriate step, show which limit law is used like in the example problem above.

5. $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x}{x^2 + 4}$

6. $\lim_{x \rightarrow \infty} \frac{7x^2 - x}{3x^2 + 2x - 8}$ (this is Problem 4 — do you get the same answer the rigorous way?)

7. $\lim_{x \rightarrow \infty} \frac{4x - 5}{2x^2 + x}$

8. Find $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 1}}{3x - 2}$ using the (a) intuition method (dropping the lower-order terms) and (b) algebraic method.

Hint: for the algebraic method, the dominant term inside the root is $4x^2$, and $\sqrt{4x^2} = 2x$ for large x . To do it by the laws, divide top and bottom by x — and remember that to move x inside the square root you write it as $\sqrt{x^2}$.

Notebook Artifact

N1. If you used the internet or AI, describe how you used it.
